AP PHYSICS 1

Unit 1 • Unit 1 Test • Kinematics • Answer key

Unit 1 Test • Kinematics

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work, start from a physics principle or equation, and justify your answers. A calculator and the AP Physics 1 equation sheet are allowed. Use $g = 10\,\text{m/s}^2$ and ignore air resistance unless stated.

Part A • Multiple choice

Question 1

Which quantity has both a magnitude and a direction?

(A) Speed
(B) Distance
(C) Acceleration
(D) Time

Solution
Answer: (C). Acceleration is a vector.

Question 2

A ball rolls 4 m east, then 7 m west. What is its displacement?

(A) 11 m east
(B) 3 m west
(C) 3 m east
(D) 11 m west

Solution
Answer: (B). $+4 - 7 = -3$ m: 3 m west.

Question 3

A car speeds up uniformly from rest to 20 m/s in 8 s. What is its acceleration?

(A) $2.5\,\text{m/s}^2$
(B) $160\,\text{m/s}^2$
(C) $0.4\,\text{m/s}^2$
(D) $12\,\text{m/s}^2$

Solution
Answer: (A). $20/8 = 2.5\,\text{m/s}^2$.

Question 4

How far does the car in the previous question travel in those 8 s?

(A) 160 m
(B) 40 m
(C) 80 m
(D) 20 m

Solution
Answer: (C). Average velocity $(0+20)/2 = 10\,\text{m/s}$, times 8 s: 80 m.

Question 5

An object has $v = +5\,\text{m/s}$ and a constant acceleration $a = -2\,\text{m/s}^2$. What is its velocity 4 s later?

(A) $+13\,\text{m/s}$
(B) $-3\,\text{m/s}$
(C) $+3\,\text{m/s}$
(D) $-13\,\text{m/s}$

Solution
Answer: (B). $v = 5 + (-2)(4) = -3\,\text{m/s}$: it has turned around.

Question 6

A rock is dropped from rest and falls for 3 s. How far does it fall?

(A) 30 m
(B) 90 m
(C) 15 m
(D) 45 m

Solution
Answer: (D). $\frac{1}{2}(10)(3)^2 = 45\,\text{m}$.

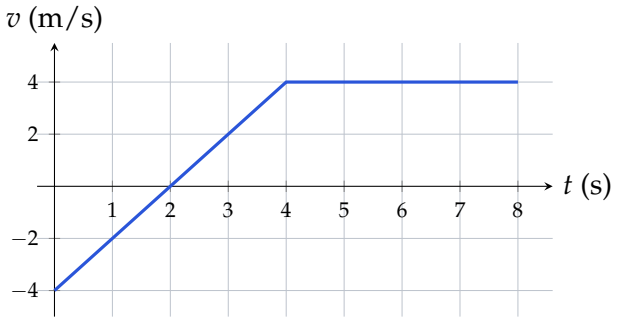
Question 7

A ball is thrown straight up at 20 m/s. What maximum height does it reach?

(A) 40 m
(B) 20 m
(C) 10 m
(D) 2 m

Solution
Answer: (B). $h = \frac{v_0^2}{2g} = \frac{400}{20} = 20\,\text{m}$.

Questions 8 to 11 refer to the graph below: the velocity-time graph of an object moving along a line.



Question 8

What is the acceleration of the object between $t = 0$ and $t = 4\,\text{s}$?

(A) 0
(B) $1\,\text{m/s}^2$
(C) $2\,\text{m/s}^2$
(D) $4\,\text{m/s}^2$

Solution
Answer: (C). Slope $= (4 - (-4))/4 = 2\,\text{m/s}^2$.

Question 9

At what time is the object momentarily at rest?

(A) 0
(B) $2\,\text{s}$
(C) $4\,\text{s}$
(D) $8\,\text{s}$

Solution
Answer: (B). $v = 0$ where the graph crosses the axis: $t = 2\,\text{s}$.

Question 10

What is the displacement from $t = 0$ to $t = 4\,\text{s}$?

(A) 0
(B) $8\,\text{m}$
(C) $16\,\text{m}$
(D) $-4\,\text{m}$

Solution
Answer: (A). Area below the axis $-4\,\text{m}$ (from 0 to 2 s) plus area above $+4\,\text{m}$ (from 2 to 4 s): 0. The object comes back to its starting point.

Question 11

What is the displacement from $t = 0$ to $t = 8\,\text{s}$?

(A) $24\,\text{m}$
(B) $32\,\text{m}$
(C) $16\,\text{m}$
(D) $20\,\text{m}$

Solution
Answer: (C). 0 for the first $4\,\text{s}$, then $4 \times 4 = 16\,\text{m}$.

Question 12

An escalator moves up at $0.5\,\text{m/s}$. A person walks up the escalator at $1.0\,\text{m/s}$ relative to the steps. What is the person's speed relative to the building?

(A) $1.5\,\text{m/s}$
(B) $0.5\,\text{m/s}$
(C) $1.0\,\text{m/s}$
(D) $2.0\,\text{m/s}$

Solution
Answer: (A). Same direction: $1.0 + 0.5 = 1.5\,\text{m/s}$.

Question 13

A ball is thrown horizontally at $12\,\text{m/s}$ from a height of $5\,\text{m}$. How far from the launch point, horizontally, does it land?

(A) $6\,\text{m}$
(B) $24\,\text{m}$
(C) $60\,\text{m}$
(D) $12\,\text{m}$

Solution
Answer: (D). $5 = \frac{1}{2}(10)t^2 \Rightarrow t = 1\,\text{s}$; $x = 12 \times 1 = 12\,\text{m}$.

Question 14

Which quantity stays constant during the flight of a projectile (no air resistance)?

(A) The vertical velocity
(B) The speed
(C) The horizontal velocity
(D) The height

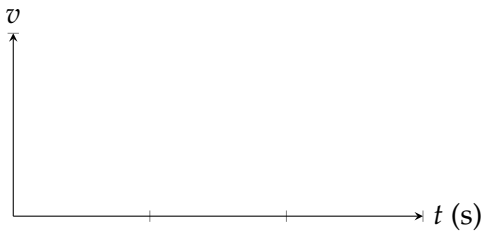
Solution
Answer: (C). $a_x = 0$, so v_x is constant. The vertical velocity, the speed and the height change.

Part B • Free response

Question 15 (Free response • Mathematical routines)

A ball is thrown straight up at 10 m/s from the edge of a roof 15 m above the ground. It misses the roof on the way down and lands on the ground.

(a) Find the time to reach the highest point and the maximum height above the ground.
(b) Find the total time from the throw until the ball hits the ground.
(c) Find the speed of the ball just before it hits the ground.
(d) Sketch the velocity-time graph of the ball (up is positive), from the throw to the impact. Label the values.



Solution

(a) $t = 10/10 = 1\,\text{s}$; it rises $\frac{10^2}{20} = 5\,\text{m}$, so $15 + 5 = 20\,\text{m}$ above the ground. (b) From the top it falls 20 m: $20 = 5t^2$, $t = 2\,\text{s}$; total $1 + 2 = 3\,\text{s}$. (c) $v = gt = 10(2) = 20\,\text{m/s}$ (or $v^2 = 10^2 + 2(10)(15) = 400$). (d) A straight line of slope $-10\,\text{m/s}^2$, from $+10\,\text{m/s}$ at $t = 0$, through 0 at $t = 1\,\text{s}$, to $-20\,\text{m/s}$ at $t = 3\,\text{s}$.

Question 16 (Free response • Experimental design and analysis)

Students want to find the acceleration of a cart rolling down a ramp. They have a meter stick and a stopwatch.

(a) Describe a procedure they could follow. Say which quantities they measure and how they reduce uncertainty.
(b) They release the cart from rest and record the time t to travel each distance d:

d (m)	0.25	0.50	0.75	1.00
t (s)	0.50	0.71	0.87	1.00

Which quantities should they plot to get a straight line? Explain using a kinematic equation.
(c) Use the data to find the acceleration of the cart.

Solution

(a) Mark several distances on the ramp; release the cart from rest at the top (no push); time the trip to each mark; repeat each measurement several times and average. (b) From rest, $d = \frac{1}{2}at^2$: plot d against t^2; the slope is $a/2$. (c) $t^2 \approx 0.25, 0.50, 0.76, 1.00\,\text{s}^2$; slope $\approx 1.0\,\text{m/s}^2$, so $a \approx 2.0\,\text{m/s}^2$.